Example 14·4

$$X_t = Z_t + \theta Z_{t-1}$$

$$= (1 + \theta B) Z_t$$

$$= \psi(B) Z_t \quad \text{where} \quad \psi(B) = 1 + \theta B$$

The spectral density function is given by

$$g(\omega) = \frac{\sigma_Z^2}{2\pi} \psi(e^{i\omega}) \psi(e^{-i\omega})$$

$$= \frac{\sigma_Z^2}{2\pi} \left[(1 + \theta e^{i\omega})(1 + \theta e^{-i\omega}) \right]$$

$$= \frac{\sigma_Z^2}{2\pi} \left[1 + \theta^2 + \theta(e^{i\omega} + e^{-i\omega}) \right]$$

$$= \frac{\sigma_Z^2}{2\pi} \left[1 + \theta^2 + 2\theta \cos\omega \right] \quad \omega \in (-\pi, \pi].$$

For $\theta = \frac{1}{2}$ $\quad g(\omega) = \frac{\sigma_Z^2}{2\pi} \left[\frac{5}{4} + \cos\omega \right]$